United States Patent Application Publication 20250279334
Kind Code A1
Publication Date September 04, 2025
Inventor(s) Wu; Banqiu

Novel 3D IC heat dissipation apparatus and method

Abstract

A self-cooled interposer for three-dimensional integrated circuit (3D IC) fabrication is disclosed. The interposer includes a thermoelectric cooler with through-cold-region vias (TCVs) that enable electrical interconnections between stacked components. A 3D chip stack comprising a graphics processing unit (GPU) and/or central processing unit (CPU), the thermoelectric-cooled interposer, and a high bandwidth memory (HBM) chip is formed, with interconnects routed through the cooled region. The self-cooled interposer effectively dissipates heat generated within the 3D IC stack, thereby significantly enhancing overall chip performance.

Inventors: **Wu; Banqiu (Katy, TX)**

Applicant: **Wu; Banqiu** (Katy, TX)

Family ID: **96880488**

Appl. No.: **19/098983**

Filed: **May 19, 2025**

Publication Classification

Int. Cl.: **H01L23/38** (20060101); **H10N10/17** (20230101)

U.S. Cl.:

CPC **H01L23/38** (20130101); **H10N10/17** (20230201);

Background/Summary

FIELD

[0001] The present disclosure generally relates to three-dimensional (3D) stacked integrated circuit

(IC) computer systems. More particularly, it pertains to a thermoelectric-cooled interposer for use in 3D IC computer systems.

BACKGROUND

[0002] In recent years, artificial intelligence (AI) technologies have emerged as some of the most competitive and transformative innovations globally. AI is expected to see widespread adoption in applications such as autonomous driving, facial recognition, expert systems, medical diagnostics, and military technology. Economically, the AI market is projected to reach a scale of up to one trillion dollars annually, driving demand for increasingly powerful AI systems.

[0003] Modern AI systems are characterized by machine learning capabilities that distinguish them from traditional modeling systems. The feasibility of AI applications largely depends on computational speed, data transfer rates, and data volume, all of which are heavily influenced by microchip performance. Advances in semiconductor manufacturing have significantly enhanced the capabilities of graphics processing units (GPUs), which now play a central role in AI systems.

[0004] The core of AI functionality-its ability to learn and train-relies on access to vast datasets, necessitating integration with large-scale data centers. As AI expands into virtually every aspect of society and industry, the demand for comprehensive AI ecosystems is set to grow significantly.

[0005] Current AI computing systems typically include GPUs, high bandwidth memory (HBM), dynamic random-access memory (DRAM), and solid-state drives (SSDs). Among these, the data transfer rate between the GPU and HBM has become the most critical factor affecting system performance.

[0006] AI system performance is strongly tied to computational capability, which in turn depends on key system metrics such as transistor gate delay, interconnect delay, and internal/external memory data transfer rates.

[0007] To address gate delay and increase memory capacity, IC feature sizes have been progressively reduced through advances in lithographic technologies. Since the 1980s, projection lithography has employed increasingly shorter wavelengths—from 436 nm (g-line) to the current 13.5 nm extreme ultraviolet (EUV)—to achieve higher IC resolution.

[0008] However, since the early 2000s, interconnect delay has become a more pressing issue than gate delay, and internal memory speed limitations—often referred to as the “memory wall”—have hindered performance. While multi-core designs in GPUs and CPUs have been adopted as a workaround, these approaches are nearing their practical limits.

[0009] To overcome these constraints, attention has shifted to 3D IC stacking, which offers the potential to significantly reduce interconnect delays and alleviate the memory wall by providing more interconnects and shorter signal paths.

[0010] HBM technology, which leverages 3D IC stacking, has already been adopted in high-end AI computing systems to improve memory transfer rates. However, the data bus between the GPU and HBM imposes bandwidth limitations due to spatial constraints. Although current buses can reach around 5,000 bits, higher bandwidths are increasingly needed.

[0011] Integration of the interposer, GPU, and stacked HBM via a data bus has been employed to enhance AI system performance. Nonetheless, traditional data bus architectures are facing geometric limitations that hinder further improvements.

[0012] Consequently, 3D integration of GPUs and HBMs is considered a promising avenue for boosting AI system performance by enabling a higher number of interconnects and reducing signal travel distance.

[0013] Despite these advantages, effective thermal management remains an unsolved challenge in 3D-integrated AI systems. Modern high-performance GPUs can consume up to 700 watts, resulting in significant heat generation that is difficult to dissipate.

[0014] As the electronics industry—one of the largest global industries—continues to push performance boundaries, IC scaling remains a key metric. However, the IEEE's international Roadmap for Devices and Systems (IRDS) forecasts that post-2027, 3D IC stacking will become

the dominant performance driver, with heat dissipation emerging as the primary technical hurdle.
[0015] The present disclosure introduces a thermoelectric-cooled interposer as an effective solution for managing heat in 3D IC stacks. This architecture allows for efficient heat extraction from the stacked area, enabling higher data transfer rates and improved thermal performance.
[0016] By integrating thermoelectric cooling technology within a 3D stacked configuration, this invention offers substantial improvements in AI computing system performance.
SUMMARY
[0017] One embodiment of the present invention discloses a thermoelectric-cooled interposer system. The thermoelectric cooler comprises a cold region, a hot region, through-cold-region vias (TCVs), a power supply, and a heat exchanger that transports heat out of the 3D stack using a coolant. Both the cold and hot regions feature thin multilayer structures.
[0018] In one embodiment, the multilayer structure consists of alternating layers of n-type semiconductor material; metal, and p-type semiconductor material. The TCVs traverse the multilayer cold region. DC circuits are present in both regions, with directional current flow determining temperature distribution. In the cold region, DC current flows from the n-type to the p-type layers; in the hot region, the direction is reversed.
[0019] The cold region is designed to be equal to or larger than the die size to ensure effective heat transfer to the hot region. The thermoelectric-cooled interposer facilitates electrical interconnection between electronic components (e.g., GPU and memory chips) on either side via the TCVs.
[0020] In another embodiment, the computer system comprises a 3D chip stack formed by a GPU chip, an HBM chip, and the thermoelectric-cooled interposer.
[0021] A key feature of the disclosed 3D IC computer system is the separation of the hot and cold regions by the thermoelectric cooler, allowing for efficient thermal management within the 3D IC stack.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0022] To facilitate understanding of the invention's features, a more detailed description is provided with reference to illustrative embodiments shown in the accompanying drawings. These drawings are intended to depict typical examples and are not to be considered limiting.
[0023] FIG. **1** illustrates an embodiment of a thermoelectric-cooled interposer system.
[0024] FIG. **2** shows an embodiment of a 3D IC computer system comprising the thermoelectric-cooled interposer and two stacked chips.
[0025] FIG. **3** presents a flow diagram illustrating one embodiment of the manufacturing method for a 3D IC stack that includes the thermoelectric-cooled interposer.
DETAILED DESCRIPTION
[0026] Embodiments of the present invention generally provide an apparatus for a 3D IC computer system. In particular, the embodiments disclosed herein relate to a 3D IC system incorporating a thermoelectric-cooled interposer.
[0027] FIG. **1** schematically illustrates a thermoelectric cooling system **100**, in accordance with one embodiment of the invention. The system **100** includes a thermoelectric-cooled interposer **102** and a cold region **104**.
[0028] In one embodiment, the interposer **102** comprises a multilayer **106** fabricated on a silicon wafer using standard semiconductor manufacturing processes such as patterning, implantation, etching, chemical vapor deposition (CVD), physical vapor deposition (PVD), chemical mechanical planarization (CMP), electrochemical deposition (ECD), thinning, and cleaning.
[0029] A through-cold-region via (TCV) **108** serves as a conductive interconnection between stacked chips. The TCV **108** has a first end **110** for establishing electrical contact with one of the

stacked chips.

[0030] The TCV **108** is a columnar conductor surrounded by a dielectric liner **112** for electrical insulation. In one embodiment, the TCV **108** is made of copper and fabricated by depositing copper into a pre-formed via hole using an ECD process.

[0031] The TCV **108** also includes a second end **114**, electrically connected to the first end **110**, enabling vertical interconnection between chips.

[0032] A direct current (DC) power supply **116** provides current through a closed loop, including conductive wires **118** and **130**, metals **120**, **124**, and **128**, an n-type silicon layer **122**, and a p-type silicon layer **126**.

[0033] The thermoelectric-cooled interposer **102** dissipates heat through a hot region **138**, which interfaces with a heat exchanger **132**. Coolant enters the exchanger via an inlet **134** and exits through an outlet **136**, thereby removing heat from the system.

[0034] As DC current flows through the loop, the cold region **104** becomes cooler than the hot region **138**. The hot region **138**, embedded in the heat exchanger **132**, transfers heat to the coolant, which then carries it out of the thermoelectric cooling system **100**.

[0035] FIG. **2** illustrates a stack system **200** comprising a GPU chip **240**, a memory chip **252**, and a thermoelectric-cooled interposer **202**. The GPU chip **240** includes an integrated circuit (IC) layer **242** and a bulk silicon layer **244**. The IC layer **242** houses semiconductor transistors and interconnects for functional operation.

[0036] The IC layer **242** is fabricated on a processor wafer using standard semiconductor processing steps such as patterning, implantation, etching, CVD, PVD, CMP, ECD, thinning, and cleaning.

[0037] In one embodiment, the IC layer **242** consists of two sub-layers: a device sub-layer with transistors and an interconnect sub-layer. Following formation, the processor wafer is thinned to prepare for stacking.

[0038] A through-silicon via (TSV) **246** passes through the bulk silicon layer **244** and is surrounded by a dielectric liner **248** for insulation. A revealed TSV end **250** provides an electrical contact point for stacking.

[0039] The TSV end **250** connects to the TCV first end **210** on the thermoelectric-cooled interposer **202**. Bonding between the GPU chip **240** and interposer **202** can be performed using traditional or hybrid bonding methods.

[0040] A TCV **208** is formed by etching a hole using high-aspect-ratio plasma etching or laser processing, followed by dielectric liner **212** deposition (via PVD or CVD) and copper deposition using ECD.

[0041] The thermoelectric-cooled interposer **202** includes a multilayer **206** composed of alternating layers of p-type semiconductor, metal, and n-type semiconductor, deposited using standard PVD or CVD techniques. Both semiconductor types are silicon-based.

[0042] On the opposite side of the interposer **202**, a second TCV end **214** connects to a contact point **258** on the memory chip **252**. In one embodiment, hybrid bonding is used to bond the interposer **202** to the memory chip **252**.

[0043] The memory chip **252** may be a dynamic random-access memory (DRAM) chip or a high bandwidth memory (HBM) chip. It comprises a device layer **254** and a bulk silicon layer **256**. The contact point **258** connects to backend interconnects on the memory chip.

[0044] A DC power supply **216** provides current through a loop that includes conductive wires **218** and **230**, metals **220**, **224**, and **228**, an n-type silicon layer **222**, and a p-type silicon layer **226**.

[0045] The stack system **200** includes a heat exchanger **232**. Coolant flows into the heat exchanger through inlet **234** and exits through outlet **236**, expelling heat from the system.

[0046] The stack system **200** features a cold region **204** and a hot region **238**. As DC current flows, the cold region **204** becomes cooler than the hot region **238**. Heat from the GPU chip **240** and memory chip **252** is transferred to the cold region **204** and then to the hot region **238**, which dissipates it through the heat exchanger **232**.

[0047] The GPU chip **240** and memory chip **252** can be stacked onto the thermoelectric-cooled interposer **202** using either traditional thermo-compression bonding or hybrid bonding at near-ambient temperatures.

[0048] FIG. **3** illustrates a 3D IC stacking flow **300** for manufacturing a 3D IC computer system, in accordance with some embodiments of the present disclosure.

[0049] The process begins at block **302**, where the processor chip **240** is fabricated on a processor wafer using standard semiconductor processing steps such as lithography, etching, cleaning, PVD, CVD, atomic layer deposition (ALD), ECD, and CMP. The TSV **246** is also formed during this step and electrically connected to the chip circuitry.

[0050] At block **304**, the bulk silicon layer **244** is thinned via grinding and planarized via CMP, exposing the TSV end **250**. The wafer is then singulated into individual processor chips **240**.

[0051] At block **306**, the thermoelectric-cooled interposer **202** is fabricated on an interposer wafer. The interposer includes a thermoelectric cooler composed of alternating materials (e.g., n-type and p-type semiconductors). When DC current flows through these materials, temperature differences arise due to differing work functions, causing one junction to heat and the opposite junction to cool. In one embodiment, a thin metal layer is placed between semiconductor layers to prevent interdiffusion. The cold and hot regions are formed from multilayer structures. TCVs are then fabricated by etching holes, depositing a dielectric liner, and filling with copper.

[0052] At block **308**, a CMP process planarizes the interposer surface and reveals the first end **210** of the TCVs in preparation for stacking.

[0053] At block **310**, the TCV first end **210** is bonded to the TSV end **250** of the processor chip **240**. In one embodiment, hybrid bonding is employed, joining both dielectric and conductive areas.

[0054] At block **312**, the interposer wafer is thinned and planarized using CMP to prepare for bonding with the memory chip **252**. The second end **214** of the TCVs is revealed during this step.

[0055] At block **314**, the memory chip **252** is fabricated. This may involve standard DRAM fabrication or the stacking of dies to form an HBM.

[0056] At block **316**, grinding and CMP processes are used to expose the connection point **258** on the memory chip **252**. Plasma treatment may be used to prepare the bonding surface.

[0057] At block **318**, the connection point **258** is bonded to the second end **214** of the TCVs on the interposer **202**.

[0058] At block **320**, the assembled wafer is singulated into individual dies using dicing methods such as sawing, laser cutting, plasma cutting, or hybrid dicing. Following singulation, the dies are packaged for final use and shipment to end users.

Claims

1. A self-cooled interposer configured to connect a first electronic component to a second electronic component, comprising: A thermoelectric cooler comprising a cold region and a hot region, wherein the cold region is configured to dissipate heat generated by the first electronic component and/or the second electronic component; and A plurality of through-cold-region vias (TCVs), wherein a plurality of first ends of the TCVs are connected to the first electronic component and a plurality of second ends of the TCVs are connected to the second electronic component.

2. The self-cooled interposer of claim 1, wherein the cold region comprises alternating layers of an n-type semiconductor, a metal layer, and a p-type semiconductor.

3. The self-cooled interposer of claim 1, wherein the hot region comprises alternating layers of a p-type semiconductor, a metal layer, and an n-type semiconductor.

4. The self-cooled interposer of claim 1, wherein the first electronic component is a graphics processing unit (GPU).

5. The self-cooled interposer of claim 1, wherein the first electronic component is a central processing unit (CPU).

6. The self-cooled interposer of claim 1, wherein the second electronic component is a memory chip.

7. The self-cooled interposer of claim 1, wherein the second electronic component is an interposer.

8. The self-cooled interposer of claim 1, wherein heat generated by the hot region is removed from the interposer via a flowing liquid coolant.

9. The self-cooled interposer of claim 1, wherein the first electronic component is bonded to the first ends of the TCVs.

10. The self-cooled interposer of claim 1, wherein the second electronic component is bonded to the second ends of the TCVs.

11. A method of stacking integrated circuit (IC) chips, comprising: a. Fabricating a first chip comprising conductive vias for interconnection; b. Fabricating a self-cooled interposer on a wafer, the interposer comprising alternating layers of an n-type semiconductor, a metal layer, and a p-type semiconductor, and including a cold region and a hot region; c. Forming a plurality of through-cold-region vias (TCVs) in the interposer and exposing first ends of the TCVs, wherein the TCV pattern aligns with the conductive vias of the first chip; d. Aligning the first chip and the interposer, and bonding the conductive vias of the first chip to the first ends of the TCVs; e. Thinning the wafer and exposing second ends of the TCVs; f. Fabricating a second chip comprising interconnect metal vias; g. Aligning the second chip and the interposer, and bonding the interconnect metal vias of the second chip to the second ends of the TCVs to form a stacked structure; and h. Singulating the stacked chips.

12. The method of claim 11, wherein the bonding in steps (d) and (g) is hybrid bonding.

13. The method of claim 11, wherein the first chip is a graphics processing unit (GPU).

14. The method of claim 11, wherein the second chip is a memory chip.

15. The method of claim 11, wherein the TCVs include dielectric liners.
